
ROADASSIST BHARAT

RoadAssist Bharat

Project Overview

| *One Platform. Every Vehicle. Every Phone. Every Road.*

Document README.md

Version 1.0 — planning baseline

Team 4 engineers · 22 weeks · 11 sprints · 18 phases

Status Baseline (frozen at Sprint 0 exit)

RoadAssist Bharat

One Platform. Every Vehicle. Every Phone. Every Road.

An offline-first, AI-powered roadside assistance and vehicle-safety platform for India — spanning cars, two-wheelers, autos, trucks, buses, tractors, and EVs, across Android, iOS, web, Android Auto, and feature phones (SMS/IVR/USSD).

Planning documents

Document	Contents
Team Charter	Team composition, RACI, rituals, sprint calendar, git/PR/commit rules, code review checklist, Definition of Done, documentation standards, quality gates, risk register
Master Roadmap	All 18 phases with objectives/tasks/dependencies/deliverables/acceptance criteria, target architecture, why each service exists, business model
D1 — Backend & Data Lead	Per-phase plan: schema (60 tables), 140 APIs, booking state machine, dispatch, auth/RBAC, sync conflict resolution, emergency engine
D2 — Frontend & Mobile Lead	Per-phase plan: design system, 48 screens, 42 components, offline client, maps, gov portal, accessibility, 8 languages
D3 — AI & Data Science Lead	Per-phase plan: 9 AI systems with inputs/outputs/algorithms/training data/metrics/baselines, on-device bundle, analytics
D4 — DevOps, QA & Security Lead	Per-phase plan: K8s/Terraform/CI-CD, observability, telecom gateway, load/chaos/security testing, deployment, DR

Team

ID	Role	Owns
D1	Backend & Data Lead	Services, PostgreSQL/PostGIS, event bus, API contracts, auth, booking/dispatch, emergency, payments
D2	Frontend & Mobile Lead	React Native app, Next.js portals, design system, offline client, maps UI, Android Auto, a11y, i18n
D3	AI & Data Science Lead	9 AI subsystems, model training/serving, on-device inference, feature store, voice/NLU, analytics
D4	DevOps, QA & Security Lead	Kubernetes, Terraform, CI/CD, observability, all testing, security, telecom gateway, compliance

Timeline

22 weeks · 11 sprints of 2 weeks · 18 phases · 10 milestones

Sprint	Weeks	Phases	Milestone
S0	1–2	Research, Planning	M0 — Baseline frozen
S1	3–4	Architecture, Database	M1 — Skeleton alive
S2	5–6	Auth, Backend APIs (I)	M2 — Login works
S3	7–8	Backend APIs (II), Frontend (I)	M3 — First booking
S4	9–10	Frontend (II), AI (I)	M4 — Vertical slice
S5	11–12	AI (II), Offline Engine	M5 — Works with no internet
S6	13–14	Maps, Vehicle Diagnostics	M6 — Diagnose & route
S7	15–16	Emergency, Gov Dashboard	M7 — Crash to responder
S8	17–18	Analytics, Testing	M8 — Quality bar
S9	19–20	Deployment, Optimization	M9 — Production
S10	21–22	Future Roadmap, GA hardening	M10 — GA + pitch

Non-negotiables

1. **Emergency paths never regress.** Two approvals, dedicated test run, staged rollout.
2. **Offline is a first-class mode, not a fallback.**
3. **Feature phones are users.** Every core journey completable over SMS/IVR.
4. **No PII leaves India.** Including logs, backups, and crash reports.
5. **We never fake it in a demo.** If it's mocked, we say so.

Status: planning baseline. No implementation code yet — S0 begins with Phase 0 research.